Gao 2021 Breast Cancer — analysis summary

RUN 3 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=8) on sample, n_pcs swept 20→40: per-neighbourhood iLISI 1.00→1.59 (of 6 samples), rising then plateauing at θ=8. kNN graph is ONE connected component at every θ (3–10); 1 detached cell removed. Patient-territorial but continuous — not an archipelago — GO.

Gate 2 — module geography → GO (non-branching)

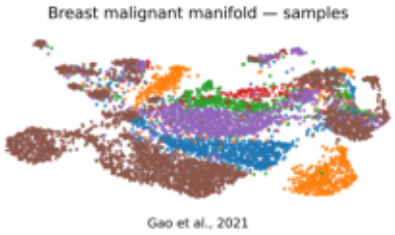
Smoothed geography: Gast↔pSom anti-correlate (r=−0.50, organized cross-repression); Crest opposes pSom (r=−0.55), weakly tracks Gast (+0.19). Along pseudotime Gast falls (ρ=−0.33) and Crest rises (ρ=+0.24; NC-high decile at t=0.77, terminal). A single Gast/pSom → Crest axis with a terminal Crest gradient — not resolved into two competing arms. Crest does not enrich a distinct Gast arm — no inversion flag.

Gate 3 — hub cross-repression → NA

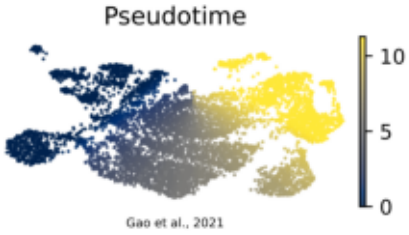
Non-branching topology. Curve: 2 tips, 0 forks (Nodes 15/20/30), fold-free (self-approach 2.7–2.9). A PPT tree returns 2 forks, reconciled as a node-density artifact, NOT a Gast/pSom Y: all arm-segments share one flat profile (Gast−pSom spread 0.009); forks dissolve as ppt_sigma rises (2→2→1→0); the 2 forks sit 2.24 UMAP units apart (median edge 0.84). No bifurcation → NA.

STEP FIGURES

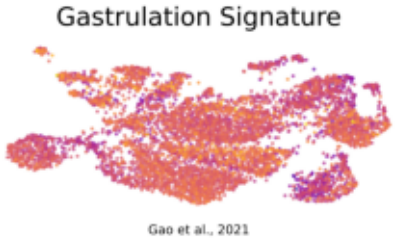
Manifold — samples



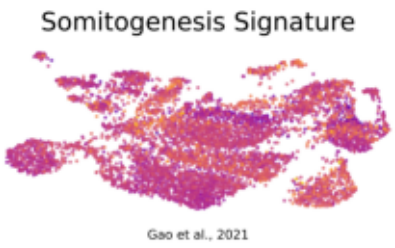
Pseudotime



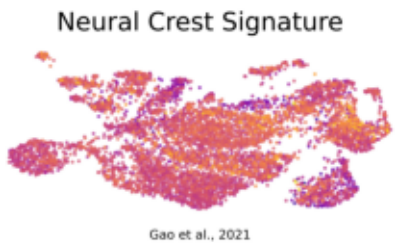
Gastrulation



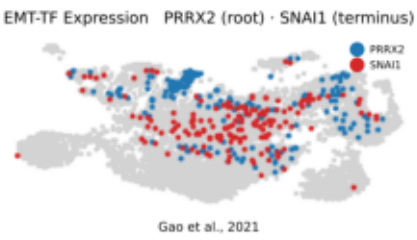
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast Cancer (Gao et al., 2021), 10x — 6,606 malignant cells (6 breast samples; thyroid ATC excluded); 6,605 after island removal.

Verdict.
The embryonic pattern reproduces as a single gastrulation/somitogenesis → neural-crest axis with organized Gast↔pSom cross-repression and a terminal Crest gradient (non-branching); the PPT tree's 2 forks are a density artifact, not a bifurcation, so Gate 3 is NA. Geometry Linear — consistent with the CCCA breast cohort. Gate 1 GO / Gate 2 GO / Gate 3 NA. Raw module means Gast −0.025 / pSom +0.020 / Crest −0.045. EMT-TFs PRRX2 (early/root) → SNAI1 (late/terminus).

Parameter choices

Threshold	100 malignant cells/sample — all 6 breast samples kept (smallest TNBC3=195); thyroid ATC excluded	**Data**	Two leaves inner-joined on 22,164 shared genes; 2,696 overdispersed; no sc.pp.scale
Embedding	Harmony θ=8 on sample; n_pcs=30, n_neighbors=30, UMAP spread=1.3	**Trajectory**	scFates curve (non-branching), Nodes=20 λ=0.3 μ=100; fold-free (SA 2.73)
Root	Gast/pSom-high, Crest-low curve tip; pseudotime vs cycling ρ=+0.40 (secondary)	**Geometry**	Linear; module means Gast −0.025 / pSom +0.020 / Crest −0.045
EMT-TFs	PRRX2 (early/root) → SNAI1 (late/terminus), detection-gated (>5%)		